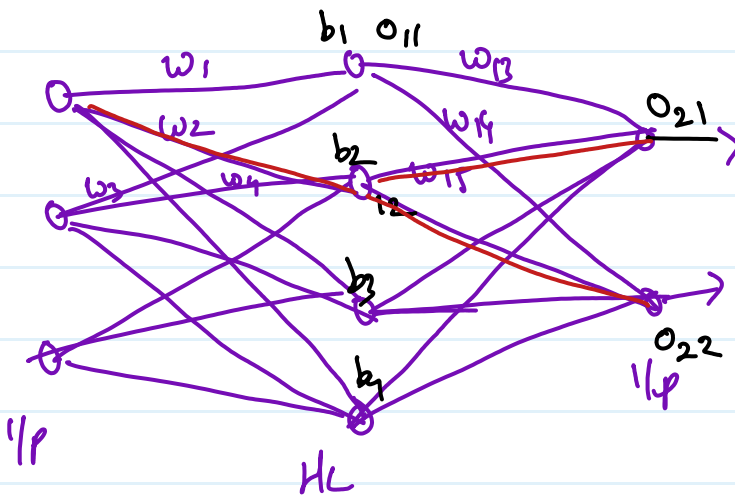


optimized

Back propagation $\rightarrow$ weight & bias update.

$$W_{\text{new}} = W_{\text{old}} - \alpha \frac{\partial L}{\partial W_{\text{old}}}$$



loss

$$W_{14 \text{ new}} = W_{14 \text{ old}} - \alpha \frac{\partial L}{\partial W_{14 \text{ old}}}$$

$$W_{2 \text{ new}} = W_{2 \text{ old}} - \alpha \left[\frac{\partial L}{\partial W_{2 \text{ old}}} \right]$$

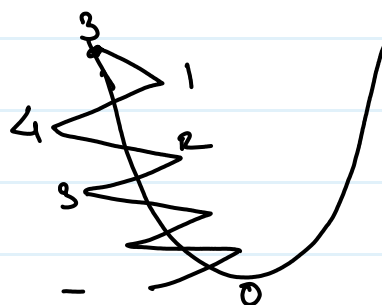
$$\frac{\partial L}{\partial W_{2 \text{ old}}} = \left[\frac{\partial L}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial O_{12}} \times \frac{\partial O_{12}}{\partial W_{2 \text{ old}}} \right] + \left[\frac{\partial L}{\partial O_{22}} \times \frac{\partial O_{22}}{\partial O_{12}} \times \frac{\partial O_{12}}{\partial W_{2 \text{ old}}} \right]$$

① Gradient Descent (GD) →

- It work on entire dataset
- To use it with small dataset
- It we required stable convergence
- It is accurate with direction
- It will slow with large dataset

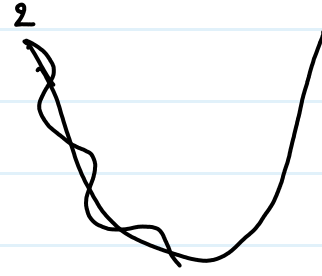
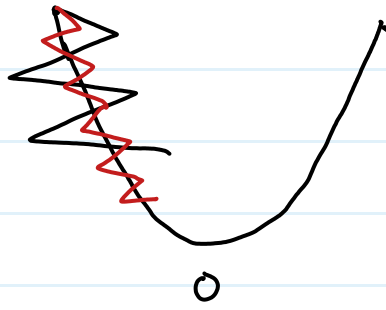
② stochastic Gradient descent (SGD)

- It work one by one on records.
- like have large dataset
- want fast update
- noise update
- Zig-zag path



③ mini batch Gradient descent

- most commonly used
- We can use it with almost every DL task.
- It is fast & stable balance.



④ SGD with momentum

- It used momentum of past gradient
- Faster with less valley.
- Best with CNN
- It is not for every DL task

⑤ Ada Grad →

x_1	x_2	x_3	x_4
1	0	0	1
0	1	0	0
0	0	0	0

- It used with sparse data
- Commonly used in RNN/NLP
- * → It increase importance of rare feature.
- With many layers LR become small.

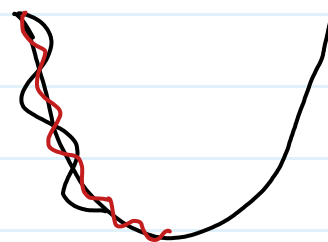
⑥ RMSprop →

- It overcome the issues of Adagrad
- It will handel learning rate before it become small.
- It use avg of few recent gradient.
- sequential data like test and time-series.

⑦ Adam optimizer*

This is the combo of SGD momentum + RMSprop.

- Faster
- Stable
- Adaptive LR
- Less turning Required
- NLP / CNN



⑧ AdamW

upgraded version of Adam optimizer.

- Better Regularization
- Used with transformers

- Used in modern DL model
- Can use with very large data.

Loss function

$$\text{loss} = \text{Actual} - \text{prediction}$$

Problem type	Loss func
① Regression	- MSE, MAE, Huber
② Binary class	- Binary cross entropy
③ multiclass	- Sparse. Categorical cross entropy
④ multi Label	- Binary cross entropy

① MSE -

→ When we have normal regression problem like price (continuous value)

→ Without outlier data or with very less noise.

② MAE →

→ To reduce impact of outlier.

③ Huber loss -

→ MSE + MAE

→ When data has outlier

→ We need stable training.

class - 4